Meilin Tong

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PLACEMENT DIRECTORS

Prof. Rui Castro rui.castro@mcgill.ca
Prof. Kensuke Sakamoto kensuke.sakamoto@mcgill.ca

EDUCATION

Ph.D. Economics, McGill University, 2021–2027 (expected)
M.Fin. Finance, Schulich School of Business, York University, 2017–2018
B.Math. Mathematics/Business Administration (Honours), Minor in Statistics, University of Waterloo (*Graduated with Distinction*), 2012–2017

WORK EXPERIENCE

Senior Business Analyst, Canadian Imperial Bank of Commerce (CIBC), 2018–2021

FIELDS OF SPECIALIZATION

Primary Research Field: Econometrics
Research Specializations: Time Series Econometrics, Macroeconometrics
Teaching Fields: Econometrics, Statistics, Finance, Macroeconomics

DISSERTATION

"Three Essays on Volatility: Modeling, Macroeconomic Identification, and High-Frequency Inference"
Committee: Prof. Jean-Marie Dufour (Chair), Prof. John W. Galbraith, Prof. Victoria Zinde-Walsh

JOB MARKET PAPER

"Estimating Higher-Order Stochastic Volatility Models with Moving-Average Components: Methodology and Applications," with Jean-Marie Dufour. September 2026, 84 pages.

Abstract: Moving-average dynamics provide a parsimonious way to capture short-run time-series patterns that may otherwise require additional autoregressive lags. Yet they are rarely incorporated into stochastic volatility models because latent volatility complicates estimation and model selection. We develop a framework for a broad class of stochastic volatility models in which latent log-volatility follows an ARMA(p,q) process, denoted SV(p,q). For this class of models, we show that appropriately transformed returns admit an ARMA representation. The proposed estimation procedure then consists of two steps: it first estimates the ARMA parameters by regression and then recovers the structural parameters through autocovariance matching. Consistency and asymptotic normality are established for the regression estimator. To guide empirical implementation, we develop a two-stage procedure for selecting the model orders. Monte Carlo simulations assess the method's finite-sample performance. In an application to daily S&P 500 returns from 1962 to 2024, the model-selection procedure favors an SV(1,1) specification, supporting the inclusion of a moving-average component in latent volatility. The empirical analysis also evaluates volatility forecasts at multiple horizons against standard GARCH-family benchmarks. Together, these contributions provide a practical framework for estimation, model selection, and forecasting with richer volatility dynamics.

WORKING PAPERS

"Gini Representations of Skewness, Kurtosis and Higher-Order Moments," with Jean-Marie Dufour and Abderrahim Taamouti. September 2026, 26 pages. Submitted to *Communications in Statistics - Theory and Methods*. (arXiv:2510.22714)

Abstract: We provide pairwise-difference (Gini-type) representations of higher-order central moments for both general random variables and empirical moments. Such representations do not require a measure of location. For third and fourth moments, this yields pairwise-difference representations of skewness and kurtosis coefficients. We show that all central moments possess such representations, so no reference to the mean is needed for moments of any order. This is done by considering i.i.d. replications of the random variables considered, by observing that central moments can be interpreted as covariances between a random variable and powers of the same variable, and by giving recursions that link the pairwise-difference representation of any moment to lower-order ones.

"From Announcement Reactions to Macroeconomic Effects: Identification-Robust Inference with Weakly Separated Policy and Information Shocks," with Jean-Marie Dufour. September 2026, 85 pages.

Abstract: Market reactions to FOMC announcements reflect both unexpected monetary-policy actions and information about the economic outlook. Empirical studies construct shock-specific proxies from high-frequency financial price movements around policy announcements, aggregate them to monthly frequency, and use them as external instruments in structural vector autoregressions to estimate dynamic macroeconomic effects. This strategy faces two identification problems: weak shock separation upstream, when announcement data do not clearly distinguish the latent shocks, and weak proxy relevance downstream, when a correctly separated proxy is weakly correlated with its target shock. Standard weak-instrument methods address the second problem, not the first. Using identification through heteroskedasticity, we show that weak separation can leave generated proxies as mixtures of structural shocks, invalidating plug-in inference, including Anderson–Rubin procedures. We develop joint inference that incorporates proxy construction directly into structural inference. A projected procedure remains asymptotically valid under weak upstream separation and weak downstream relevance, while a reduced-profile procedure is sharper under additional conditional identification and regularity conditions. Monte Carlo experiments show severe overrejection by plug-in methods, while the joint procedures avoid this distortion in the designs considered. In an FOMC application, extending the measurement window from the policy statement to include the press conference appears to sharpen inference about central-bank-information shocks when the estimated proxies are treated as fixed. Once proxy-construction uncertainty is incorporated, however, the corresponding confidence sets become unbounded, and most dynamic macroeconomic effects remain imprecisely determined. Broader announcement coverage therefore need not deliver more precise inference about macroeconomic transmission.

CONFERENCE PROCEEDINGS

"Pairwise Difference Representations of Central Moments: Skewness, Kurtosis, and Higher-Order Recursions," with Jean-Marie Dufour and Abderrahim Taamouti. October 2025, 15 pages. *2025 Joint Statistical Meetings (JSM) Proceedings*. (DOI: 10.5281/zenodo.17379223)

WORK IN PROGRESS

"Simulation-Based High-Frequency Jump Tests," with Md. Nazmul Ahsan and Jean-Marie Dufour.

"High-Dimensional Stochastic Covariance Estimation via Model Aggregation," with Md. Nazmul Ahsan and Jean-Marie Dufour.

TEACHING EXPERIENCE

Teaching Assistant, McGill University
ECON 304 Financial Instruments and Institutions, 2024–2026
ECON 353 Macroeconomics Honours 1, 2025
ECON 302 Money, Banking and Government Policy, 2023–2025
ECON 257D1/D2 Economic Statistics Honours, 2021–2023
ECON 338 Introductory Econometrics 2, 2022

Teaching Assistant, University of Waterloo, 2014–2016
MATH 127 Calculus 1 for the Sciences
MATH 104 Introductory Calculus for Arts and Social Science

RESEARCH EXPERIENCE

Research Assistant, Prof. Francesca Carrieri, McGill University, 2025–present
Reconstructed historical Canadian equity indices from firm-level records dating back to 1860; conducted data cleaning and empirical analysis in R.

FELLOWSHIPS AND AWARDS

SSHRC Doctoral Fellowship, 2025–2026 (Award value: C$40,000)
FRQSC Doctoral Research Scholarship, 2025–2027 (Award value: C$33,334)
Graduate Excellence Award, McGill University, 2021–2026
Brecher Fellowship in Economics, McGill University, 2024
Award for the Best Third-Year Paper, McGill University, 2024
Dean's Honour List, Master of Finance, York University, 2018
Chartered Financial Analyst (CFA) Scholarship, York University, 2017
Math/ELAS Scholarship, University of Waterloo, 2013
University of Waterloo President's Scholarship, 2012
Faculty of Mathematics Entrance Scholarship, University of Waterloo, 2012

CONFERENCE AND SEMINAR PRESENTATIONS

"Estimating Higher-Order Stochastic Volatility Models with Moving-Average Components: Methodology and Applications," with Jean-Marie Dufour.

- CIREQ Financial Econometrics Conference (poster), Montréal, September 2026.
- 18th Annual Society for Financial Econometrics (SoFiE) Conference, Macau, June 2026.
- 20th CIREQ–CIRANO Ph.D. Students' Conference, Montréal, May 2026.
- CIREQ Lunch Seminar, Montréal, February 2026.
- CIREQ Weekly Ph.D. Student Research Workshop, Montréal, January 2026.
- 59th Canadian Economics Association (CEA) Annual Meeting, Montréal, May 2025.

"Pairwise Difference Representations of Moments: Gini and Generalized Lagrange Identities," with Jean-Marie Dufour and Abderrahim Taamouti.

- Joint Statistical Meetings, American Statistical Association, Nashville, August 2025.

"From Announcement Reactions to Macroeconomic Effects: Identification-Robust Inference with Weakly Separated Policy and Information Shocks," with Jean-Marie Dufour.

- CIREQ Lunch Seminar, Montréal, July 2026.
- 60th Canadian Economics Association (CEA) Annual Meeting, Vancouver, May 2026.
- CIREQ Lunch Seminar, Montréal, May 2026.

PROFESSIONAL SERVICE

Referee, *Journal of Quantitative Economics*
Discussant, Canadian Economics Association Annual Meeting, 2025
Innovation Commissioner, Post-Graduate Students' Society, McGill University, 2025–2026

LANGUAGES

English (fluent), Mandarin (native), French (basic)

TECHNICAL SKILLS

MATLAB, R, Python, SQL, Stata, LaTeX

CITIZENSHIP

China

REFERENCES

Prof. Jean-Marie Dufour	McGill University	jean-marie.dufour@mcgill.ca
Prof. John W. Galbraith	McGill University	john.galbraith@mcgill.ca
Prof. Victoria Zinde-Walsh	McGill University	victoria.zinde-walsh@mcgill.ca